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Linux Admin  
Continue set up  
Ubuntu server

1.

Use the command **Cat /etc/apt/sources.list** to list repositories in the Ubuntu server. Pictured below:

```
ms3213@UbuntuServer1:~$ cat /etc/apt/sources.list
# Ubuntu sources have moved to /etc/apt/sources.list.d/ubuntu.sources
ms3213@UbuntuServer1:~$ cat /etc/apt/sources.list.d/ubuntu.sources
Types: deb
URIs: http://us.archive.ubuntu.com/ubuntu/
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
```

<https://www.baeldung.com/linux/default-repositories-debian-ubuntu>

<https://help.ubuntu.com/community/Repositories/CommandLine>

```
DESCRIPTION
  add-apt-repository is a script which adds an external APT repository to either /etc/apt/sources.list or a file in /etc/apt/sources.list.d/ or removes
  an already existing repository.
```

**sudo add-apt-repository** can be used to install official repositories. To install PPAs or personal package archives, you must use **sudo add-apt-repository ppa:user/ppa**. The ppa and user can be found on <https://launchpad.net/ubuntu/+ppas>

```
-P, --ppa
  Add an Ubuntu Launchpad Personal Package Archive.

  Must be in the format ppa:USER/PPA, USER/PPA, or USER.

  The USER parameter should be the Launchpad team or person that owns the PPA. The PPA parameter should be the name of the PPA; if not provided, it defaults to 'ppa'.

  The GPG public key of the PPA will also be downloaded and added to apt's keyring.

  To add a private PPA, you must also use the --login parameter, and of course you must also be subscribed to the private PPA.
```

The above screenshot shows the format that the command has to be in to add a PPA

Source: **man add-apt-repository** to get detailed help from the Ubuntu command line  
<https://help.ubuntu.com/community/Repositories/CommandLine>

```
ms3213@UbuntuServer1:~$ sudo add-apt-repository universe
[sudo] password for ms3213:
Sorry, try again.
[sudo] password for ms3213:
Adding component(s) 'universe' to all repositories.
Press [ENTER] to continue or Ctrl-c to cancel.^CAborted.
ms3213@UbuntuServer1:~$ sudo add-apt-repository universe
Adding component(s) 'universe' to all repositories.
Press [ENTER] to continue or Ctrl-c to cancel.
Hit:1 http://us.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:3 http://us.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:5 https://ppa.launchpadcontent.net/orchestra/ppa/ubuntu noble InRelease
Err:6 https://ppa.launchpadcontent.net/orchestra/ppa/ubuntu noble Release
      404 Not Found [IP: 185.125.190.80 443]
Reading package lists... Done
```

The above picture shows the command installing the universe repository successfully using the **sudo add-apt-repository** command.

```
ms3213@UbuntuServer1:~$ sudo apt install tmux
[sudo] password for ms3213:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
tmux is already the newest version (3.4-1ubuntu0.1).
tmux set to manually installed.
2. 0 upgraded, 0 newly installed, 0 to remove and 34 not upgraded.
```

The command **sudo apt install tmux** installs tmux. Typing **tmux** then opens it, and it looks like below. Tmux allows you to open multiple terminals in a single window. To exit out of it, press **ctrl b d** on the keyboard.

In the screenshot below, tmux is running, showing a new terminal session inside the current terminal window.



<https://www.shells.com/en-US/tutorial/Installing-and-using-tmux-on-Ubuntu-20-04>

```
File Edit Options Buffers Tools Help
Welcome to GNU Emacs, one component of the GNU/Linux operating system.
To follow a tip, click Mouse-1 on it, or move to it and type M-1.
To quit a partially entered command, type Control-g.

Important Help menu items:
Emacs Tutorial      Learn basic Emacs keyboard commands
Read the Emacs Manual  View the Emacs manual using Info
Noninteractive       GNU Emacs comes with absolutely NO WARRANTY
Copying Conditions    Conditions for redistributing and changing Emacs
Where Manuals / Ordering Manuals How to order printed manuals from the FSP

Useful tasks:
Visit New File       Specify a new file's name, to edit the file
Open Some Directory  Open your home directory, to operate on its files
Customize Startup    Change Initialization settings including this screen

GNU Emacs 29.5 (build 1, x86_64-pc-linux-gnu, GTK+ version 3.24.41,
built on 2024-11-04) of 2024-10-03, modified by Debian
Copyright (C) 2024 Free Software Foundation, Inc.
```

<https://www.gnu.org/software/emacs/tour/>  
<https://www.pluralsight.com/resources/blog/cloud/how-to-install-emacs>  
[https://www.reddit.com/r/emacs/comments/n4rpw2/help\\_how\\_do\\_i\\_exit\\_emacs\\_from\\_the\\_terminal/](https://www.reddit.com/r/emacs/comments/n4rpw2/help_how_do_i_exit_emacs_from_the_terminal/)

The second screenshot, it shows that it successfully downloaded

In picture 4, I ran `sudo fail2ban status`, which shows us what is being monitored to ban in this case, we have 1 jail protecting sshd. So any Ip addresses trying to get into ssh will be banned.

```
ms3213@UbuntuServer1:~$ sudo apt install fail2ban
[sudo] password for ms3213:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-pyasyncore python3-pyinotify whois
Suggested packages:
  monit sqlite3 python-pyinotify-doc
The following NEW packages will be installed:
  fail2ban python3-pyasyncore python3-pyinotify whois
0 upgraded, 4 newly installed, 0 to remove and 34 not upgraded.
Need to get 496 kB of archives.
After this operation, 2,572 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

```
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ms3213@UbuntuServer1:~$
```

```
ms3213@UbuntuServer1:~$ systemctl status fail2ban.service
● fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: enabled)
   Active: active (running) since Mon 2025-11-10 01:55:17 UTC; 9min ago
     Docs: man:fail2ban(1)
   Main PID: 5427 (fail2ban-server)
    Tasks: 5 (limit: 2268)
  Memory: 35.7M (peak: 36.0M)
     CPU: 1.937s
   CGroup: /system.slice/fail2ban.service
           └─5427 /usr/bin/python3 /usr/bin/fail2ban-server -xf start
```

```
ms3213@UbuntuServer1:~$ sudo fail2ban status
[sudo] password for ms3213:
sudo: fail2ban: command not found
ms3213@UbuntuServer1:~$ sudo fail2ban-client status
Status
|- Number of jail:      1
  - Jail list:  sshd
```

By default, the SSH jail is enabled, by default and any other jail you would want has to be enabled.

<https://www.digitalocean.com/community/tutorials/how-to-protect-ssh-with-fail2ban-on-ubuntu-20-04>

3. Here, we **run `sudo apt install cowsay`** to install the utility. Then, to test the utility, we run **`cowsay woooooow`** or any other text we want the cow to say, and it will print an image using ASCII characters of a cow saying the text. **Man `cowsay`** tells the user how to use it. The first picture below shows the installation, and the second shows it being run with the text “**woooooow**”.

```
ms3213@UbuntuServer1:~$ sudo apt install cowsay
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  filters cowsay-off
The following NEW packages will be installed:
  cowsay
0 upgraded, 1 newly installed, 0 to remove and 34 not upgraded.
Need to get 10.6 kB of archives.
After this operation, 93.2 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu noble/universe amd64 cowsay all 3.03+dfsg2-8 [10.6 kB]
Fetched 10.6 kB in 0s (76.8 kB/s)
Selecting previously unselected package cowsay.
(Reading database ... 146211 files and directories currently installed.)
Preparing to unpack .../cowsay_3.03+dfsg2-8_all.deb ...
Unpacking cowsay (3.03+dfsg2-8) ...
Setting up cowsay (3.03+dfsg2-8) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

```
ms3213@UbuntuServer1:~$ cowsay [Wooooooooooooooooooooooooooooooooooooooooooooowwwwwwwwwwww]
 _____
/ [Wooooooooooooooooooooooooooooooooooooooooooooowwwwwwwww \ 
\ wwwww ]                                     /
-----
      /\
     /(oo)\_____ )\\
    (__)\       )\\/
        ||----w |
         ||     ||
```

Here we install lol cat by running **sudo apt install lolcat**. You can run lol cat by putting lolcat and then the name of a file to output it in rainbow text. In this case, I did lolcat greplab, which printed the whole file in rainbow.

```
ms3213@ubuntuServer1:~$ sudo apt install lolcat
[sudo] password for ms3213:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  fonts-lato javascript-common libs-jquery library-libruby3.2 rake ruby-net-telnet ruby-optimist ruby-paint ruby-rubygems ruby-sdbm ruby-webrick
  ruby-xmhrpc ruby3.2-rubygems-integration unzip zip
Suggested packages:
  anac2 | lighttpd | httpd ri ruby-dev bundler
The following NEW packages will be installed:
  fonts-lato javascript-common libs-jquery library-libruby3.2 lolcat rake ruby-net-telnet ruby-optimist ruby-paint ruby-rubygems ruby-sdbm ruby-webrick
  ruby-xmhrpc ruby3.2-rubygems-integration unzip zip
0 upgraded, 15 newly installed, 0 to remove and 54 not upgraded.
Need to get 9,315 kB of archives.
After this operation, 42.2 MB of additional disk space will be used.
```

The above picture shows the lolcat being run on the Greplab file. Displaying in rainbow colors.  
<http://tecmin.com/lolcat-color-output-linux-terminal/>

To install Vim, we run **sudo apt install vim**

For this server it tells us that Vim is already up to date

Vim is a text editor similar to Emacs. To use it, type **vim**, then the name of the file. This will open the file and let you navigate, search through and edit text.

```
ms3213@UbuntuServer1:~$ sudo apt install vim
[sudo] password for ms3213:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
vim is already the newest version (2:9.1.0016-1ubuntu7.9).
vim set to manually installed.
4. 0 upgraded, 0 newly installed, 0 to remove and 34 not upgraded.
```

The above shows how to install vim.



The above shows the Vim editor open with my test file, hellotest, run with the command **vim hellotest**. To exit without saving, type:q! And to save and exit run :wq

<https://opensource.com/article/19/3/getting-started-vim>

<https://askubuntu.com/questions/420364/how-do-i-install-and-get-started-with-vim-vi>

**Grep** `install /var/log/dpkg.log` lists installation dates for installed packages. The **whereis** command can be used to locate the folder of a package.

```
tmux installed on 4/11/2026    /usr/bin/tmux
emacs installed on 4/11/2026   /usr/bin/emacs
fail2ban installed on 4/11/2026 /etc/fail2ban
cowsay installed on 4/11/2026  /usr/games/cowsay
lolcat installed on 4/11/2026  /usr/games/lolcat
vim updated on 4/11/2026      /usr/bin/vim
```

<https://unix.stackexchange.com/questions/12578/list-packages-on-an-apt-based-system-by-installation-date>

<https://serverfault.com/questions/250937/finding-where-something-is-installed-on-ubuntu>